Landmark proximity corridor for the Open Sea

Nov 8, 2026 Arjun about 90 seconds

One-minute script

> The Open Sea is the big empty square after the right turn, about 40 metres long. There is no wall next to the rover, so our wall-following has nothing to hold on to. GPS was 6 to 15 metres off here, and the gyro slowly drifts, so a rover that only trusts those two would wander.
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> So we steer on things that do not move. Two of them are plain walls: the B wall behind us and the end wall ahead. In the middle are the landmarks: a tree with its structure, the two poles of the campus store, and the store front. In the picture, the grey dots are what the LiDAR saw, and the blue circles are those landmarks.
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> The rover measures how far the poles are to its left and keeps that distance, using a small controller we call the proximity PID. The distance to keep, 6.65 metres, comes from the teaching run. Later, the store front at 3.0 metres takes over. The B wall behind and the end wall ahead keep the heading straight.
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> It works: in two full runs, log 8 and campus 9, the rover crossed from wall to wall with no hold, no pause and no failsafe, and its heading changed by under 6 degrees over more than 40 metres. In simulation the worst case is 0.41 metres off, before it was 1.02. Still open: a strong gyro drift gives about 0.4 metres of error at the end.

What you need to understand

Why landmarks: in the Open Sea the LiDAR map matching has no walls to match. A landmark is a fixed object we can measure our side distance to, so every scan corrects the position by itself. It does not depend on GPS or on how long we have been driving.

- The blue circles are the LiDAR points of the tree and its structure, the two campus-store poles and the store front. The LiDAR sees shapes, not names, so the box fits a straight line through the points it sees there and measures the distance to that line.
- The proximity PID: error = measured distance minus the target (6.65 m). The correction is kp·error + ki·sum of errors + kd·change of error, with 10, 1.5 and 20, limited to ±8 degrees and changed slowly (0.5 degrees per scan), so the heading failsafe never sees a jump.
- The poles give only the side position. Their direction wobbles by about ±6 degrees, so the heading comes from the straight walls (B wall behind within ±1.3 degrees in six runs).
- Nine logs were measured with the same functions as the firmware: the pole-line distance was 6.5 to 6.8 m in five runs (teach run 6.67 m) and was seen in 39 to 56 percent of the scans.

Likely questions

Why not use GPS? In log 10 the GPS was 6 to 15 metres off next to the buildings, even with good satellite numbers. Walls and poles do not move.

What if a person stands in front of a pole? The box only uses the line when enough points fit it. If not, it keeps the last good heading for a while. The people simulation (next slide) covers crowds.

Is it tested on the real rover? Two real autonomous crossings (log 8, campus 9) and a closed-loop simulation. Version 10 has not been driven yet.